

MINING

- Exhausts the ores thus it is considered as robber industry.
- Cannot be restored once the mine is exploited.
- Minerals are formed by geological process over a long period of time.

Factors affecting exploitation of minerals:

- Distribution
- Value of mineral & metal
- Mining cost
- Grade of ore
- Size of the deposit
- Transport cost
- Labour

Impact of mining:

- Employment
- Transportation development
- Migration of population for employment
- Huge earnings from exports
- Excess usage leads to exhaust of mines
- Decrease water table & pollutes environment

IRON

- Constitutes 5% of earth's crust
- Widely distributed
- Can be recycled to produce steel (decentralized iron industry)

In India, Major producing states are Orissa, Jharkhand, Chhattisgarh, Karnataka, Goa, Andhra Pradesh and Tamil Nadu. Rich Hematite in Singhbhum region with support industries in Damodar basin.

Types of iron ore:

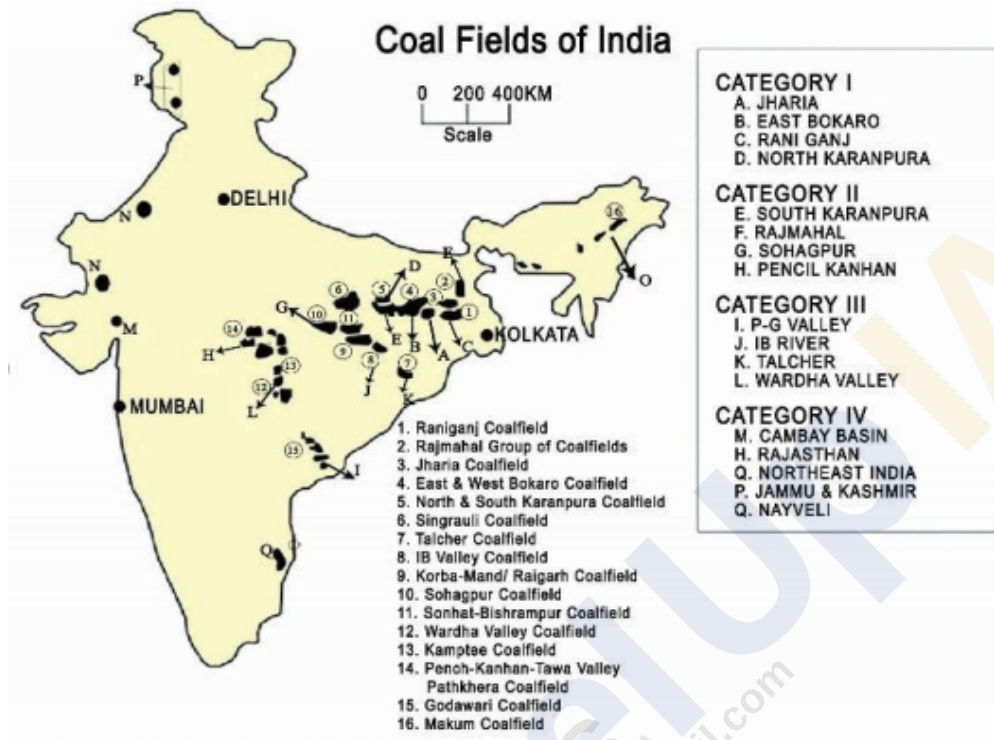
Magnetite, hematite, limonite, siderite



COAL:

- India faces severe coal shortage.
- Formed by compressed vegetative matter
- Indian is world's 2nd largest producer & consumer of coal
- The India Energy Outlook 2021 report of the International Energy Agency (IEA) said energy use in India has doubled since 2000, with 80% of demand still being met by coal, oil and solid biomass.
- According to the IEA's Coal Report 2021, India's coal consumption will increase at an average annual rate of 3.9%.

Coal types & features →
taught in class



Issues/challenges:

- High reliance on coal for energy production.
- Rise of energy demands in India.
- Cheaper & abundant availability
- Availability of other renewable resources.
- High economic cost.

PETROLEUM

- Derived from decaying marine organic matter
- Found in sedimentary rocks of various ages.

Medium of transporting:

- Road – oil trucks
- Rails – tank wagons
- Sea – oil tankers
- Aircraft – military / emergency purpose

Pipelines for transporting oil

Transported to refineries or coastal shipping terminals

Fixed route

Most economical way of transporting

Problems:

Continuous maintenance

Corrosive in nature

Leakage may exist

Political turmoil

URANIUM

Ores: Uranyl, pitch blende

Major deposits in India: Jaduguda, Tumalapalle, Bhatin, Narwapahar, Turamdih



ELECTRICITY

The principal source of energy.

It is inexhaustible

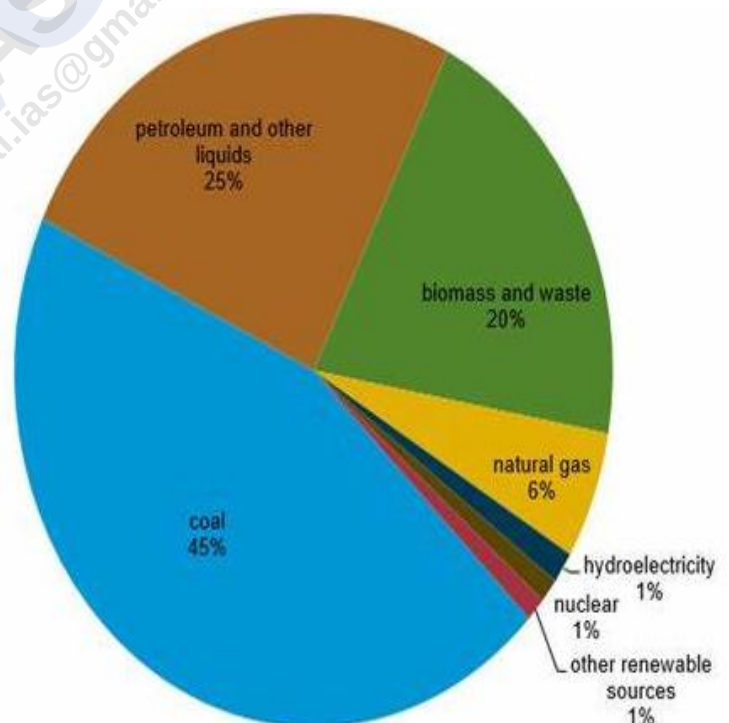
Cleaner energy

Highly adaptable

Easily transferred but for short distances

Important for industries

Cannot be stored



Solar potential of India

Himalayan region has very high potential of developing solar energy.

Advantages	Disadvantages
New and Renewable resource	Fluctuating wind speeds alter the production projections.
No GHG emissions, water pollution.	Threats to avian fauna.
Cheapest among all renewable resource	Create noise pollution
Very easy to install and get started	Land use competition (Agriculture, Housing)
	Often built on remote locations.

Wind energy of India

India's wind power potential at 120 meters height is estimated to be 695 GW. Current, installed wind power in India is 38 GW.

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Nuclear Plant

India stands at 7th position in terms of the number of nuclear reactors, with over 23 nuclear reactors in 7 power plants across the country.

Located in isolated areas

Initial cost is high

Low running cost,

low cost of fuel transportation,

more efficient than steam power plant, requires highly skilled labour, less standby losses.

COPPER

Brown in colour

Found in igneous & metamorphic rocks

Found in metal state

Major producers of India: Singhbhum, Hazaribagh, Khetri, Arguncha-rampura in Bhilwara

LEAD & ZINC

Soft & malleable

One of the earliest metals use by humans

Lead → glass making, medicine, sheets & pipes

Zinc → Coating, galvanizing iron, alloys

Major producers in India: Rajasthan, Gujarat, Andhra Pradesh

GOLD & SILVER

Gold is Corrosion free.

Gold is used as Bullion, currency, jewellery

Silver tarnishes due to contact with sulphur

Resistant to Acetic acid

Ore: Argentite

Major producers of gold in India: Karnataka, Andhra Pradesh, Assam, Bihar, MP, Tamil Nadu

Major Producers of silver in India: Zawar mines, leads smelters, kolar, Hutti

OIL SEEDS

India is one of the major oilseeds grower and importer of edible oils.

The self-sufficiency in oilseeds attained in India through "Yellow Revolution" during early 1990s

India accounts for about 15-20 per cent of global oilseeds area.

Major producers in India are Rajasthan, Gujarat, Tamil Nadu, Madhya Pradesh, Haryana, Maharashtra, Karnataka, Andhra Pradesh.

Growing condition for oilseeds:

20-30 degree Celsius

50-75 cm rainfall

Sandy loams, red,
yellow or black soil

Susceptible to frost &
drought, rain

MILLETS

2018 observed as the 'National Year of Millets'

Millets are also known as nutria-cereals.

- Sorghum (Jowar)
- Pearl Millet (Bajra)
- Finger Millet (Ragi)

With stand high temperature and drought

Can be grown in poor soils and difficult terrains

Anti acidic, gluten free

Helps prevent type 2 diabetes

Bioethanol can be created

United Nations has declared the year 2023 as the International Year of Millets

Major producers in India : Rajasthan, Andhra Pradesh, Telangana, Karnataka, Tamil Nadu, Maharashtra, Gujarat, U.P and Haryana.

Growing condition

26-29 degree Celsius

450-650 cm rainfall
(Sorghum)

50-60cm (kodo millet)

40—45 cm (finger
millet)

Sandy loams, alluvial,
loamy soil

Initiatives by Govt.:

Integrated cereals
development programs

Nutritional security
through intensive millet
promotion (Rastriya
Krishi Vikas Yojana)

Rainfed area
development program

RICE

One of the very few crops which can sustain excess of heat as long as there is sufficient water
Machines should be robust enough to work in flooded fields.

Major producers in India: UP, West Bengal, AP, TN, PJ, OD

Growing condition

20-27 degree Celsius

200-250 cm rainfall

Heavy clayey, loamy
soil

COFFEE

Expensive than tea

Contains caffeine

Three major species : Arabica, Robusta, Liberica

Major producers in India: Karnataka, Kerala, TN, AP, Od, Tripura, Nagaland,

Growing condition

Not <11 degree Celsius

100-200 cm rainfall

Well drained volcanic soil

FISHING

Fish concentration depends on Temperature, depth of water, ocean currents, salinity which determines the amount of fish food or planktons.

Overfishing and wasteful fishing needs to be checked.

Fish is Source of essential minerals

India has rich inland water resources

Major fishing ground across world: NW pacific region

NE Atlantic waters

NW Atlantic region

NE Pacific

SE pacific

West central Pacific

Oceans jurisdiction → explained in class

LIVESTOCK

Issues in tropics:

Climate & vegetation

Cattle breed

Diseases

Traditional practices

Capital intensive

Forest resources & lumbering → explained in class

GROUND WATER IN INDIA

The water that seeps through rocks and soil and is stored below the ground

